Sources	Deficiency Symptoms	Toxicity
Dietary, excellent sources; citrus fruits and their juices, sweet red and green peppers, strawberries, cantaloupe, broccoli, Brussels sprouts, papaya, cauliflower. Good sources: potatoes, tomatoes, peas, kale, asparagus, cabbage.	Scurvy: fatigue, bleeding gums, poor appetite, slow wound healing, muscle fatigue. Seen occasionally in infants fed only cow's milk and in elderly people	Relatively nontoxic; supplement abuse may cause diarrhea
Dietary, excellent sources: pork, sunflower seeds, fortified grain products, fresh peas. Good sources: legumes, whole and enriched grains	Beriberi: poor coordination, Muscle weakness, edema, heart diseases	Nontoxic except for high dose injections.
Dietary, excellent sources: liver, milk and milk products Good sources: oysters, mushrooms, green vegetables (broccoli, spinach, asparagus)	Ariboflavinosis: inflammation of the mouth and tongue, cracks at the corners of the mouth and lips, anemia, dermatitis, eye – related problems	Nontoxic
Dietary, excellent sources: tuna, chicken, liver, beef, fortified cereals. Good sources: Mushrooms, whole grains	Pellagra: diarrhea, dermatitis, dementia, and possibly death (the four Ds) common in Africa and Asia	Nicotinic acid form is toxic with high dose supplements. Symptoms include flushing (>100 mg); liver damage, increased levels of uric acid and glucose in blood (>3 grams); increased use of glycogen during exercise.
Dietary, excellent sources: liver, potatoes, bananas, prune juice, light poultry meat, fortified grain products Good Sources : Sweet potatoes, water – melon	Convulsions, dermatitis, inflamed tongue, and anemia. Deficiency symptoms often seen in alcoholics, the elderly, and women taking birth control pills	Toxic with long – term supplement abuse (>1 gram per day). Symptoms include nerve damage and lack of muscular control (usually reversible)
Dietary, excellent sources: legumes, liver spinach and turnip greens, artichokes. Good sources : asparagus, oranges, green peas, broccoli, beets, beet greens, green beans, corn	Macrocytic megaloblastic anemia, diarrhea, mental confusion, depression, fatigue.	Nontoxic
Dietary, excellent sources: organ meats, shellfish, milk and milk products (found only in animal foods)	Pernicious anemia: a macrocytic megaloblastic anemia accompanied by nerve damage. Most commonly results from poor absorption due to lack of intrinsic factor, not from low dietary intakes of vitamin	Nontoxic
Dietary, excellent sources: liver, soy flour, egg yolks. Dietary, good Sources: cereals. Nondietary: synthesized by bacteria in the intestinal tract.	Poor appetite, nausea, sore tongue, depression, pallor, hair loss, dry skin, increased blood levels of cholesterol and bile Rare in humans.	Nontoxic
Dietary, excellent sources: egg yolks, liver, kidney. Dietary, good sources: meat, legumes, whole grains	Deficiency symptoms are rare. In severely malnourished individuals symptoms such as paresthesia in the toes and soles of feet, burning sensations in the feet, fatigue, insomnia, depression may be seen	Nontoxic

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